

SolarSmart Homes Sell Faster

The housing industry may be in a slump in parts of the country, but in and around Sacramento, California, one type of housing is selling at twice the rate of its competitors—the SolarSmart Home. These solar-powered, super-energy-efficient new homes are the brainchild and brand of the local Sacramento Municipal Utility District (SMUD). To earn the SolarSmart designation, home builders must install a checklist of cost-effective energy efficiency upgrades, combined with solar generation. Started in 2007, the brand is meeting with surprising acceptance from local builders. With agreements to build 1,900 homes so far, SolarSmart homes will constitute about 35%—and maybe more—of all new homes built in SMUD territory in 2008.

“If we had gotten 15% of new homes [to meet SolarSmart qualifications], we would have felt great,” says Wade Hughes, SMUD’s SolarSmart new-homes program manager, “but we are doubling that. Now we think 60% of new homes might be a better goal.” Hughes attributes his program’s success to its ease—just 12 required measures (see Table 1). He says builders prefer a prescriptive approach with simple instructions, rather than messing around trying to meet an energy savings target. Simple as these qualifications are, they still produce major energy savings—savings of 31% to 38% more than the state’s already strict energy code, Title 24. When the electricity produced by the built-in PV is factored in, these homes are also helping SMUD to manage peak demand (see “Zero Peak Homes: Real Homes, Real Savings,” p. 34). Cumulatively, the 1,900 SolarSmart homes are expected to shave nearly 3 megawatts off peak demand on hot summer days.

Unlikely as it might seem at first, the slump in the housing market has

helped to jump-start the program’s success. “Current market conditions are favorable to the SolarSmart program, because builders are looking to distinguish themselves,” says Hughes. While local builders enjoyed long lines at grand openings in recent years, those days are now gone. Builders now have to compete to get home buyers’ attention and dollars. One home builder who at first dismissed the program’s requirements as too expensive, called back three months later to sign on. “He told me he was not only in the business of building homes, but he also had to sell them,” says Hughes, “and the SolarSmart brand provided a significant marketing advantage.”

The program also provides a sliding scale of incentives for builders. Those builders who took the most risk and signed on earliest received the largest incentives—about \$9,000 per home. Today’s incentives are closer to \$6,000 per home.

For consumers, the cost of purchasing a SolarSmart home is not much more than the cost of a traditional home in price per square foot, according to Hughes. He says that builders are absorbing the added costs of installing the efficiency upgrades—about \$12,000 per home after all of the incentives—and chalking those costs up to marketing fees. Comparing the upgrade costs and benefits to other marketing efforts, why not? Builders could be spending \$45,000 to buy one double-page spread in the local newspaper’s Sunday edition, but what seems to be on most home buyers’ minds these days is energy efficiency—at least according to the many home buyer surveys that SMUD has conducted. And the better educated the consumer is, the more likely he or she is to want a home that saves energy, according

Table 1. Required SolarSmart Measures

Measure	Base	SolarSmart Homes
Attic insulation	R-38	R-38
Radiant barrier	No	Yes
Wall insulation	R-13	R-13 + 1 in R-4.2 foam
Quality installation*	No	Yes
Low air infiltration*	No	Yes
Windows	Energy Star	Energy Star
Furnace AFUE	0.80	0.90+
A/C	13 SEER	14 SEER/12 EER
ACCA design*	No	Yes
Duct testing*	No	Yes
Fluorescent	Meets Title 24	All recessed fixtures
Solar PV	NA	2kW AC solar system

*HERS-verified upgrades

to Hughes. That’s not surprising, he says, given that SolarSmart homes save the home buyer money from the first month. “Even if the builder passed on all the costs of installing the solar and energy efficiency upgrades, the customer would be revenue positive every month.”

In the Sacramento area, summer days routinely hit 100°F and above, so the efficiency upgrades are focused on saving cooling energy use. Most of these cooling savings come during SMUD’s peak hours, when SMUD has to buy power on the open market to meet its customers’ peak demands. That power is the most expensive for SMUD and comes from the dirtiest sources. By shaving peak demand, these homes benefit not only their owners, but also all SMUD customers in two different ways. First, decreased peak demand translates to fewer greenhouse gas emissions, which helps SMUD meet state-mandated greenhouse gas reduction targets. Second, since SMUD may not have to spend money on building new plants, customers’ rates are stabilized. With so many benefits to be had, it’s no wonder that Hughes thinks home builders soon may have to work harder to explain to customers why they should even consider buying a home that isn’t SolarSmart.

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